# Blood: source-bounded personal anxiety scoring from glucose and wearable signals

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**Blood is a private personal health record that turns automatic CONTOUR NEXT ONE glucose readings and phone-derived health signals into one visible stabilization score, one learned time-block pattern, and six aligned graphs: anxiety, glucose, heart rate, heart-rate variability (HRV), sleep, and steps. The anxiety estimate is not a diagnosis and does not infer mental state from a model trained on other people. It is a deterministic, source-bounded index calculated from the latest uploaded glucose, heart rate, HRV, sleep, and step states. The score starts from a neutral personal baseline, adds positive weight for physiologic states that can feel like stress or reduce recovery, subtracts weight for steady source states, scales the result to a one-decimal 1–10 display, and selects one positive food, water, or movement action from the strongest high, low, short, light, or raised source state. The pattern window groups recent source records by Eastern time block and predicts which block has been more unstable from the same bounded factors, recalculating as new uploads arrive. The anxiety graph reconstructs historical score points from stored glucose and health samples, then appends the latest score as the newest point. When Health Connect does not expose true RMSSD HRV, Blood marks HRV as estimated and computes it only from independent, dense, low-noise sleep or resting heart-rate windows. A live read on June 27, 2026 at 6:13 PM ET produced a 4.4/10 score from 82 mg/dL glucose, 68 bpm heart rate, 12 ms estimated HRV, 431 minutes asleep, and 13,028 recent steps. The system is therefore a transparent self-monitoring surface: every score, pattern, and graph can be reconstructed from the source values and rule table.**

# 1 Introduction

Many personal health apps show glucose, sleep, heart rate, or activity as separate traces. That separation is technically clean, but it leaves the user to decide which signal explains the current body state. Blood was built for the opposite interface problem: a current glucose reading or wearable signal should not become another small dashboard burden. It should become one clear score, then the source graphs that make the score auditable.

The score is intentionally called an anxiety estimate rather than an anxiety diagnosis. Blood does not classify psychiatric anxiety, does not use population diagnosis labels, and does not claim that one biomarker proves a mental state. The implementation uses physiologic signals only as personal stabilization evidence. Low blood glucose can produce adrenergic symptoms such as fast heartbeat, shaking, sweating, nervousness, or anxiety (*1*). HRV standards define variability from beat-to-beat RR intervals and distinguish true HRV measurement from looser heart-rate summaries (*2*). Reduced HRV is associated with anxiety disorders in meta-analytic literature, but the effect is group-level and not sufficient for diagnosing an individual state (*3*). Blood therefore treats HRV as one recovery signal among several, not as a standalone mental-health label.

The data path is also bounded. Android Health Connect exposes categories such as activity, sleep, vitals, and blood glucose, and Samsung Health can share selected Samsung Health data through Health Connect (*4*, *5*). In the current Blood deployment, glucose reaches the server from a Blood Bridge path connected to the CONTOUR NEXT ONE meter, while heart rate, sleep, and steps come from Health Connect records written by the phone or wearable. The bridge reads Health Connect records in pages, and the interface separates health-upload time from Samsung/Health Connect source time, so a recent bridge upload cannot make an older shared heart-rate sample look current. When a source does not provide true HRV, Blood computes a labeled proxy from heart-rate samples only when the sample geometry is strong enough to make the estimate useful.

# 2 Results

## 2.1 A single score remains reconstructable from source values

Blood renders the anxiety estimate first, then a learned instability window, and then six aligned graphs for anxiety, glucose, heart rate, HRV, sleep, and steps. The placement matters because the score summarizes the latest uploaded source state, the pattern window names the time block that has carried more source instability, and the graphs are the audit trail. The display value is computed by:

$$S = \text{round}_{0.1}\left(\min\left(10, \max\left(1, 2R\right)\right)\right)$$

where $R$ is the raw additive score and $S$ is the visible 1–10 score. The raw score starts at $R = 2.2$. Source-specific factors then add or subtract the values shown in Table 1. Positive values raise the anxiety estimate. Negative values lower it.

**Table 1.** Deterministic anxiety-score factors in the current Blood deployment. The final visible score is $2R$, clamped to 1–10 and rounded to one decimal place.

| Signal | Rule | Raw change |
|---|---|---:|
| Glucose | <70 mg/dL | +1.00 |
| Glucose | >180 mg/dL | +0.90 |
| Glucose | <82 or >140 mg/dL | +0.35 |
| Glucose | 82–140 mg/dL | -0.15 |
| Heart rate | ≥100 bpm | +0.85 |
| Heart rate | ≥85 bpm | +0.40 |
| Heart rate | 55–75 bpm | -0.15 |
| HRV or estimated HRV | <25 ms | +0.80 |
| HRV or estimated HRV | <40 ms | +0.45 |
| HRV or estimated HRV | ≥65 ms | -0.25 |
| Sleep | <300 minutes asleep | +0.90 |
| Sleep | <360 minutes asleep | +0.45 |
| Sleep | ≥420 minutes asleep | -0.25 |
| Steps | <1,500 recent steps | +0.35 |
| Steps | <4,000 recent steps | +0.15 |
| Steps | ≥8,000 recent steps | -0.25 |

This additive design is deliberately small. A glucose crash, very short sleep, high heart rate, or low HRV can move the score, but no single signal silently becomes the whole answer. Stable signals can offset a stressed signal. Missing signals do not create fake precision; they simply do not contribute.

## 2.2 The live score has a full audit trail

Table 2 reconstructs the live Blood score checked on June 27, 2026 at 6:13 PM ET. The latest score is not a black-box output. It is the arithmetic sum of five visible source states.

**Table 2.** Live score reconstruction from the Blood summary API on June 27, 2026 at 6:13 PM ET.

| Term | Source state | Raw contribution |
|---|---|---|
| Baseline | Personal neutral starting point | +2.20 |
| Glucose | 82 mg/dL, inside the target band | -0.15 |
| Heart rate | 68 bpm, inside the calm band; source sample 1:50 PM ET | -0.15 |
| Estimated HRV | 12 ms estimated HRV, below 25 ms; estimate timestamp 4:04 AM ET | +0.80 |
| Sleep | 431 minutes asleep, at least 420 minutes | -0.25 |
| Steps | 13,028 recent steps, at least 8,000 | -0.25 |
| Raw total | 2.20 – 0.15 – 0.15 + 0.80 – 0.25 – 0.25 | 2.20 |
| Visible score | 2 × 2.20, clamped and rounded | 4.4/10 |

The label associated with the score is also deterministic: scores at or below 2.5 are labeled low, scores through 4.5 steady, scores through 6.5 watch, scores through 8 high, and higher scores very high. The live 4.4/10 score is therefore labeled steady.

## 2.3 HRV is source-preferred and proxy-labeled

The HRV value used in the anxiety score is true HRV only when a source HRV/RMSSD record exists. Blood preserves that value and labels its basis as source RMSSD. When a true HRV record is absent, Blood estimates HRV from heart-rate samples and labels the result with `ms_est`. The estimator converts heart rate to an RR interval using:

$$RR_i = \frac{60000}{\mathrm{bpm}_i}$$

It then computes an RMSSD-like proxy over adjacent RR differences:

$$\widehat{HRV} = \sqrt{\frac{1}{n}\sum_{i=2}^{n+1}(RR_i - RR_{i-1})^2}$$

The current estimator rejects weak windows before producing a value. It deduplicates heart-rate records by timestamp, prefers points inside sleep windows when at least 18 such points exist, otherwise searches resting windows, and evaluates 30-minute candidate windows. A candidate must contain at least 18 points, at least 14 accepted adjacent pairs, a duration of at least 18 minutes, median sample gap no higher than 2.25 minutes, coverage at least 80%, median heart rate no higher than 92 bpm, heart-rate standard deviation no higher than 7 bpm, median adjacent heart-rate step no higher than 3.5 bpm, 90th-percentile adjacent heart-rate step no higher than 8 bpm, and 90th-percentile adjacent RR difference no higher than 160 ms. Blood removes noisy adjacent pairs with median-absolute-deviation gates before computing the RMSSD-like value. It then selects independent or low-overlap candidate windows, requires at least two selected windows and 40 accepted pairs, and reports the median value across the selected windows.

This is stronger than a day-wide heart-rate swing, but it is still not the same as a chest-strap or ECG-grade beat-to-beat HRV record. The deployed API therefore carries `basis`, `quality`, `confidence`, sample count, accepted-pair count, rejected-pair count, median gap, coverage ratio, selected-window count, and window length. The live value is 12 ms estimated from sleep heart-rate samples with 735 samples, 240 accepted pairs, eight selected windows, sleep-dense estimate quality, and highest-available-without-beat-intervals confidence; its source timestamp is 4:04 AM ET, not a live RMSSD feed.

# 3 Discussion

Blood's anxiety score is best understood as a personal stabilization index. It is useful when it reduces the work of deciding which current body signal deserves attention. It is not useful if it is treated as a medical diagnosis, a universal anxiety scale, or proof that one biomarker explains a mood state.

The main design choice is transparency. The score can be computed by hand from the table. The front page shows the score first because it is the latest uploaded source state Alan asked to see, and it places the pattern window and graphs immediately after the score because those are the source audit. The anxiety graph uses the same formula as the top score and the same x-axis as the source graphs, so changes in the score can be compared directly with glucose, HR, HRV, sleep, and steps. The same structure also keeps recommendations bounded. Blood selects the next suggestion from the strongest positive factor, not from a generated personality profile. The wording is deliberately simple but not vague: source reason, high/low/short/light/raised state, and one positive eating, drinking, or movement action. Blood recommendations do not restrict exercise or use negative restriction wording. If glucose is too low or near the low edge, Blood names the glucose value and asks for a carb-plus-protein snack and water. If glucose is too high or near the high edge, it asks for protein/fiber with carbs, water, and an easy walk. If heart rate is too high or raised, it names the bpm value and asks for water, a protein/fiber snack, and an easy walk. If estimated HRV is too low, it names the estimated HRV value and asks for water, protein/fiber meal rhythm, and a gentle walk. If sleep is too short or light, it names the sleep duration and asks for protein/fiber with the first meal, water, and easy movement. If steps are too low or light, it names the recent step count and asks for water, normal meals, and short walks. The action label is "Latest upload", not a live moment state, because the phone bridge and Health Connect upload can lag behind the page view. The separate pattern window uses time blocks only to identify the block that has recently carried more source instability.

The strongest limitation is HRV. HRV is defined from beat-to-beat intervals, and sampled heart-rate records are a compressed derivative of that signal (*2*). Blood reduces that problem by selecting dense sleep/rest windows and labeling the proxy, but the result remains a proxy. The second limitation is source freshness. A watch can monitor heart rate continuously while the health bridge has not uploaded recently, or while Samsung Health has not exposed the newest shared sample through Health Connect. Blood treats those as separate stale states and shows both the health-upload time and the sample timestamp rather than inferring a newer value. A direct Samsung Health current-feed path would require Samsung's proprietary Health Data SDK or a separate watch-sensor bridge, not only the current Health Connect phone bridge. The third limitation is calibration. The current weights are personal, deterministic, and inspectable; they are not learned from a clinical dataset and have not been validated against symptom diaries. A future version should store Alan's own reported anxiety, caffeine, meal, medication, illness, and stressor context when he chooses to provide it, then evaluate whether the fixed weights should be changed.

# 4 Materials and Methods

## 4.1 System inputs

The Blood server stores glucose readings and health metrics separately. Glucose records include measured time, capture time, source, reading identifier, mg/dL value, relation to meal when available, and specimen source. The current automatic glucose path is the CONTOUR NEXT ONE meter bridge. Health metrics include heart rate, HRV when exposed, sleep sessions, and step windows. Blood Bridge requests HRV permission when available, but missing HRV permission does not block heart-rate, sleep, or step uploads. Its recurring background worker reads a recent two-day Health Connect window and follows Health Connect pagination so dense Samsung heart-rate records can reach the Blood API in upload-sized batches. The public summary endpoint is `/api/blood/summary`; write endpoints require the Blood ingest token.

## 4.2 Score implementation

The current anxiety score is implemented by `estimateAnxietyState` in `server.js`. The function receives the latest glucose, heart-rate, HRV, sleep, and recent-step values from the summary builder. It initializes `raw = 2.2`, applies the factor table, computes `score = clamp(1, 10, raw * 2)`, and rounds to one decimal place. It returns `score`, `scale`, `label`, `raw`, `factors`, `suggestion`, and the note "Personal stabilization estimate from latest uploaded health signals; not a diagnosis."

The anxiety trend is implemented by `estimateAnxietyTrend`. It walks stored glucose readings and health-trend samples, reconstructs score points from the most recent source values available at each timestamp, and requires at least two source inputs before keeping a historical point. It also appends the latest score as the newest visible point when a score exists. This makes the top score and the anxiety chart share one formula while keeping the graph tied to source records rather than generated filler.

The time-pattern surface is implemented by `estimateInstabilityPatterns`. It takes recent glucose readings plus health trends, assigns each source record to an Eastern time block, scores each record with the same source-specific high, low, short, light, or raised factors used by the latest score, and reports the block with the highest recent average instability. Dense heart-rate samples are reduced to hourly median buckets so a watch sampling frequently cannot dominate the pattern only by producing more rows. The reported pattern contains the current block, predicted block, source counts, observations, directional condition text, and a basis line that states the 45-day rolling window. Because the summary endpoint recomputes this object from stored source records, the pattern changes automatically when additional glucose, HR, HRV, sleep, or step data changes the distribution.

### 4.3 HRV estimator

The HRV trend function first keeps any source HRV records and marks them as source RMSSD. It then calculates proxy HRV only for dates without source HRV. Heart-rate records are grouped by date, deduplicated by `measuredAt`, sorted by time, and converted to RR intervals. Candidate windows are continuous 30-minute spans with no accepted sample gap above 2.5 minutes. Windows with too few samples, too few accepted adjacent pairs, short duration, sparse coverage, high median heart rate, high heart-rate standard deviation, high adjacent heart-rate steps, or high RR-difference noise are rejected. The selected estimate is the median RMSSD-like value from independent or low-overlap candidates, with metadata exposed in the API and chart tooltip.

## 5 Acknowledgements

The paper records the current Blood implementation and Alan Pham's requested interface contract: the anxiety score belongs at the top of the page, formatting must remain compact and consistent, the six aligned graphs follow immediately, chart values carry source timestamps, the header names health-upload freshness versus Samsung/Health Connect source freshness, and HRV must be labeled according to the strongest source basis available.

## 6 Funding

No external funding is associated with this software paper.

## 7 Author contributions

Alan N. Pham defined the application need, source preferences, scoring placement, source-freshness requirement, and automatic-bridge requirements. Codex implemented the current paper and checked it against the deployed Blood source.

## 8 Competing interests

The author declares no competing interests.

## 9 Data availability

The live public summary exposes current compact Blood state at the Blood summary API. Private write tokens, raw export tokens, and private device data are not public.

## 10 Code availability

The implementation is in the Blood application source, including `server.js`, `app.js`, and the Android bridge connector. Public routes are `https://blood.aolabs.io/` and the fallback mirror `https://aolabs.io/blood/`.

## 11 Additional information

This paper is a single-user system record. It should not be used for diagnosis, treatment, or medication decisions. The anxiety estimate is a personal software index for stabilizing attention around current physiologic signals.

## References

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